Summary

The electricity system is undergoing a period of immense change in its technologies and operation. The U.S. Department of Energy's (DOE's) Office of Electricity requested an Issue Paper from the National Academies of Sciences, Engineering, and Medicine that elevates and synthesizes insights for creating effective and efficient research, development, and demonstration (RD&D) in the electricity system. The authors present promising specific areas for innovation and the structural elements needed to enhance the effectiveness and efficiency of the electricity RD&D system. The discussion of these elements is structured into four categories: the locus of electricity innovation and which actors are producing these advances, the connectivity and integration of the innovation system and how to overcome its gaps, opportunities for partnerships and collaborations, and how to attain leverage for creating the largest innovation impacts with a set amount of resources. The U.S. electricity system is at a pivotal point in its evolution. DOE can engage with its innovation partners in different ways throughout the technology life cycle to create an electricity system that is future-ready and serves the public interest.

Introduction

RD&D efforts are essential underpinnings for the electricity system's ability to support the livelihoods of its users and enable the growth and strength of the U.S. economy. RD&D endeavors produce new technologies, services, and operational capabilities that enter into use throughout the system with the rate of diffusion affecting how quickly the electric sector can change.

Often, the RD&D process is characterized as being top-down (driven by organizational leadership) or bottom-up (coming from researchers and employees). There is an increasing recognition over the past two decades that the top-down and bottom-up processes are intertwined (Li 2023; Owen 2009). In this Issue Paper the authors recognize the role and value of both. Top-down processes set the architectural and directional dimensions of the ecosystem and bottom-up processes leverage diverse capital, knowledge, and resource bases that give shape and context to top-down RD&D aspirations.

Supporting an efficient and effective innovation ecosystem for electric power will be important for unlocking the capabilities to address the challenges and opportunities facing this system. At the time of writing, these include strains on system reliability (DOE 2025a; NERC 2025), challenges to grid resilience (Costello 2019; Gibson 2020; NASEM 2017), accelerating demand growth (ACP 2025; ICF 2025), concerns over electricity prices (McGee and Tyndall 2025; Wiser et al. 2025), regulatory paradigms that may unnecessarily constrain innovation (Kiesling et al. 2025; Schwartz 2022), and increasingly complex supply chain considerations (Howard 2025). The U.S. federal government has a strong history of supporting RD&D endeavors for critical technologies and systems, with DOE's efforts specifically enabling improvements in reliability and affordability for the U.S. energy systems (Urecki and Das 2025). These outcomes result not just from DOE's direct research work but also from its engagement of a broad network of innovation partners including national laboratories, universities, and the private sector (NASEM 2021a).

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